

Name of Student:

Sch. No.:

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

DEPARTMENT OF ELECTRICAL ENGINEERING

END TERM EXAMINATION May.-2024, B. Tech. IV Semester Electrical

Sub: Generation of Electrical Power (E-224)

Time: 3.00 Hours

Max. Marks: 50

Note: Assume missing data if necessary. Both parts of a question should be solved at ONE place only.

I.	a) With neat diagram explain the Electricity market models in deregulated power systems.	05	CO1
	b) A power station is to supply three region of load whose peak loads are 20MW, 15MW and 25 MW. The annual load factor is 50% and the diversity factor of the load at the station is 1.5. Determine following: i) Maximum demand on the station ii) Installed capacity iii) Annual energy supplied	02 01 02	CO1
	a) Discuss the sinking fund and fixed percentage methods to determine the depreciation charges.	05	CO2
	b) A bank of capacitors each 40 μ F is used to improve the power factor of a 3- ϕ , 400V, 40A load at 0.8 lagging power factor. Find the overall power factor and the ratio of new line loss to original line loss if the capacitors are connected in (a) star (b) delta.	05	CO2
III.	a) Draw the neat diagram of SCADA in thermal power stations and explain its working.	05	CO3
	b) Discuss the criterion that has to be kept in view while selecting the site for thermal power stations.	05	CO3
IV	a) With the help of a neat sketch describe the working of a fast breeder reactor used in a nuclear power plant.	05	CO4
	b) Explain with neat diagram the working of pumped storage plant and discuss its advantages.	05	CO4
V	a) Derive the short term Hydro-Thermal coordination equations using penalty factors.	05	CO5

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	b) The incremental costs of two units are $\frac{dF_1}{dP_1} = 0.25P_1 + 50 \text{ Rs/MW-hr}, \quad \frac{dF_2}{dP_2} = 0.15P_2 + 60 \text{ Rs/MW-hr}$ Where P_1 and P_2 are outputs in MW. The system load varies as under: 12 midnight to 6 AM : 60 MW 6 AM to 9 PM : 200 MW 9PM to 12 midnight : 60 MW The capacity of each unit is 150MW. (i) Would it be more economical to keep both units operating during the whole period of 24 hours or shut down one unit during low load period. A cost of Rs. 200 is incurred in taking a unit off and returning it to service (ii) Find economic loading during the high load period? (iii) Find the total fuel cost for the 24 hour period if loads are allocated as per economic scheduling.	CO5	
01			
02			
02			

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